

Group 2

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The Kmap for Binary to Decimal Converter is given below:

Decimal	A	B	C	D	L ₁	L ₂	L ₃	L ₄	L ₅	L ₆	L ₇	L ₈	L ₉	L ₁₀
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	0	0	0	0	0	0	0	0	0
2	0	0	1	0	1	1	0	0	0	0	0	0	0	0
3	0	0	1	1	1	1	1	0	0	0	0	0	0	0
4	0	1	0	0	1	1	1	1	0	0	0	0	0	0
5	0	1	0	1	1	1	1	1	1	0	0	0	0	0
6	0	1	1	0	1	1	1	1	1	1	0	0	0	0
7	0	1	1	1	1	1	1	1	1	1	1	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1	1	0	0
9	1	0	0	1	1	1	1	1	1	1	1	1	1	0
10	1	0	1	0	1	1	1	1	1	1	1	1	1	1

$$L_2 = A + BC + BD$$

$$(010)0 + A =$$

$$L_7 = A + B$$

For L1:

AB \ CD	00	01	11	10
00		1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$L1 = A + B + C + D$$

For L2:

AB \ CD	00	01	11	10
00			1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$L2 = A + B + C$$

For L3:

AB \ CD	00	01	11	10
00			1	
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$L3 = A + B + CD$$

For L4:

AB \ CD	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

$$L4 = A + B$$

For L5:

AB \ CD	00	01	11	10
00				
01		1	1	1
11	1	1	1	1
10	1	1	1	1

$$L5 = A + BC + BD$$

$$= A + B(C + D)$$

For L_6 :

AB \ CD	00	01	11	10
00				
01			1	1
11	1	1	1	1
10	1	1	1	1

$$L_6 = A + BC$$

For L_7 :

AB \ CD	00	01	11	10
00				
01			1	
11	1	1	1	1
10	1	1	1	1

$$L_7 = A + BCD$$

For L_8 :

AB \ CD	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

$$L_8 = A$$

For L_9 :

AB \ CD	00	01	11	10
00				
01				
11	1	1	1	1
10		1	1	1

$$L_9 = AB + AC + AD$$

$$= A(B + C + D)$$

For L_{10} :

AB \ CD	00	01	11	10
00				
01				
11	1	1	1	1
10			1	1

$$L_{10} = AB + AC$$

$$= A(B + C)$$

